

The Spacetime Accumulation Model (STAM): A Unified Accumulation-Field Framework for Gravitation, Propagation, and Cosmological Distance Phenomenology

Sean Brady

drwho5535@gmail.com

2 May 2026

Abstract. The **Spacetime Accumulation Model** (STAM) introduces a dimensionless scalar field, $\mathbf{A}$, termed the *accumulation field*, intended to provide a single organizing variable for several phenomena commonly treated separately: gravitational acceleration, gravitational redshift and distance–redshift relations, signal propagation delay, and horizon formation. In STAM, matter–energy acts as a source of accumulation; spatial gradients of $\mathbf{A}$ define the local gravitational acceleration, while line integrals of $\mathbf{A}$ along null trajectories define propagation and traversal effects. A critical condition, $\mathbf{A} = 1$, is postulated to mark a transition to horizon-like behavior.

1. Conceptual Postulates

1. **Existence of an accumulation field.** Spacetime is characterized (in addition to standard geometric structure) by a dimensionless scalar field $\mathbf{A}(\mathbf{x})$ representing cumulative “spacetime accumulation.”
2. **Source principle.** Matter–energy sources accumulation such that, in the spherically symmetric weak-field limit, $\mathbf{A}$ reduces to a dimensionless potential proportional to the Schwarzschild radius divided by distance.
3. **Local dynamics from gradients.** The observed gravitational acceleration field is identified operationally with the spatial gradient of $\mathbf{A}$ (up to a constant factor), rather than positing gravity as a primitive force law.
4. **Propagation from path integrals.** Propagation delay and redshift–distance “excess” are attributed to path-integrated accumulation along trajectories.
5. **Critical threshold.** The condition $\mathbf{A}=1$ defines a critical boundary associated with horizon-like propagation behavior.

The core claim is that matter and energy generate spacetime accumulation. Local gradients of this accumulation field produce gravitational acceleration. Path-integrated accumulation produces traversal delay, redshift–distance excess, and apparent luminosity–distance stretching. At a critical threshold, accumulation reaches unity, corresponding to horizon-like behavior.

2. Local (Spherically Symmetric) Limit

In the local, static, spherically symmetric case, STAM defines the accumulation field as a dimensionless function of radius:

$$\mathbf{A}(r) = \mathbf{R}_s / r$$

where the Schwarzschild radius is defined by:

$$\mathbf{R}_s = 2\mathbf{GM} / c^2$$

Equivalently:

$$\mathbf{A}(r) = 2\mathbf{GM} / (c^2 r)$$

STAM posits that local gravitational acceleration is given by the gradient of accumulation:

$$\mathbf{g} = (c^2/2) \nabla \mathbf{A}$$

In the weak-field spherical limit, this identification recovers the Newtonian inverse-square law:

$$g = GM / r^2$$

Accordingly, STAM treats “gravity” as an emergent observational description of motion within an accumulation gradient, rather than as a primitive interaction specified independently of $\mathbf{A}$.

3. Propagation Effects and Critical Threshold

STAM further assigns propagation effects to accumulated traversal through $\mathbf{A}$. For a light signal propagating along a path segment ds in the vicinity of a mass distribution, the model defines an accumulation-induced delay of the form:

$$\Delta t = (1/c) \int \mathbf{A}(r) ds$$

Substituting $\mathbf{A}(r) = 2\mathbf{GM}/(c^2 r)$ yields:

$$\Delta t = (2\mathbf{GM}/c^3) \int (ds/r)$$

which reproduces the standard weak-field logarithmic dependence associated with the Shapiro time-delay form. In STAM language, the delay is attributed to increased accumulated traversal along the signal path.

A horizon-like condition is represented by the critical threshold:

$$\mathbf{A} = 1$$

In the spherical expression $\mathbf{A}(\mathbf{r}) = \mathbf{R}_s / \mathbf{r}$, the threshold implies:

$$\mathbf{R}_s / \mathbf{r} = 1$$

and therefore:

$$\mathbf{r} = \mathbf{R}_s$$

STAM interprets this condition as a critical accumulation boundary. For $\mathbf{A} < 1$, outward propagation is modeled as remaining in an ordinary regime; at $\mathbf{A} = 1$, the model posits a transition in propagation behavior consistent with a horizon criterion; for $\mathbf{A} > 1$, outward propagation is taken to enter a trapped or otherwise state-changed regime. (The precise dynamical characterization of the $\mathbf{A} > 1$ regime constitutes an explicit open problem.)

4. Cosmological Distance Phenomenology

At the cosmological level, STAM proposes an observational decomposition in which an inferred distance can be written as the sum of (i) a geometric-distance proxy and (ii) a traversal “excess” attributed to path accumulation:

$$\mathbf{D}_{\text{adj}}(\mathbf{z}) = \mathbf{D}_{\text{geo}}(\mathbf{z}) + \mathbf{D}_{\text{excess}}(\mathbf{z})$$

The proposed geometric-distance proxy is parameterized (phenomenologically) as:

$$\mathbf{D}_{\text{geo}}(\mathbf{z}) = \mathbf{L} \mathbf{z} (1 + 0.15 \mathbf{z})$$

and the adjusted (luminosity-distance-equivalent) form is written as:

$$\mathbf{D}_{\text{adj}}(\mathbf{z}) = \mathbf{L} \mathbf{z} (1 + 0.5 \mathbf{z}) + \mathbf{b} \mathbf{z}$$

where $\mathbf{L} = \mathbf{c}/\mathbf{H}$ is a Hubble-length scale and $\mathbf{b}$ is introduced as a catalog/path calibration parameter.

$$\mathbf{D}_{\text{excess}}(\mathbf{z}) = \mathbf{b} \mathbf{z} + 0.35 \mathbf{L} \mathbf{z}^2$$

STAM then identifies this excess with a path-integrated accumulation contribution. Defining the average path accumulation by:

$$\langle \mathbf{A}_{\text{path}} \rangle(\mathbf{z}) = \mathbf{D}_{\text{excess}} / \mathbf{D}_{\text{geo}}$$

which (for the parameterizations above) becomes:

$$\langle \mathbf{A}_{\text{path}} \rangle(\mathbf{z}) = (\mathbf{b}/\mathbf{L} + 0.35 \mathbf{z}) / (1 + 0.15 \mathbf{z})$$

A corresponding local (differential) path accumulation is defined as:

$$\mathbf{A}_{\text{path,local}}(\mathbf{z}) = \mathbf{d}(\mathbf{D}_{\text{excess}}) / \mathbf{d}(\mathbf{D}_{\text{geo}})$$

which yields:

$$\mathbf{A}_{\text{path,local}}(\mathbf{z}) = (\mathbf{b}/\mathbf{L} + 0.70\mathbf{z}) / (1 + 0.30\mathbf{z})$$

$$\mathbf{D}_{\text{adj}}(\mathbf{z}) = \int [1 + \mathbf{A}_{\text{path,local}}(\mathbf{z})] \mathbf{dD}_{\text{geo}}$$

With these definitions, the adjusted distance can be expressed as:

This constitutes STAM's central cosmological phenomenological claim: the empirically inferred redshift–distance behavior may be decomposable into a shorter geometric proxy plus an additional traversal contribution that is representable as an accumulation integral. In this reading, the “unadjusted” curve corresponds to the geometric proxy $\mathbf{D}_{\text{geo}}$, while the “adjusted” curve corresponds to an observational (luminosity-distance-like) mapping that includes the traversal excess $\mathbf{D}_{\text{excess}}$. A key requirement for theoretical completeness is an independent derivation of the proposed $\mathbf{A}_{\text{path}}(\mathbf{z})$ law from physical microstructure or large-scale spacetime dynamics, rather than from curve-fitting alone.

5. Operational Inference: Mass from Accumulation Proxies

Because $\mathbf{A}$ is intended as a unifying descriptor, STAM frames a family of mass estimators as operationally equivalent projections of the same underlying accumulation strength (subject to regime-appropriate approximations):

$$\text{Horizon radius: } \mathbf{M} = \mathbf{c}^2 \mathbf{r}_h / (2\mathbf{G})$$

$$\text{Acceleration: } \mathbf{M} = \mathbf{g} \mathbf{r}^2 / \mathbf{G}$$

$$\text{Orbital velocity: } \mathbf{M} = \mathbf{v}^2 \mathbf{r} / \mathbf{G}$$

$$\text{Shapiro coefficient: } \mathbf{M} = \mathbf{K} \mathbf{c}^3 / (2\mathbf{G})$$

$$\text{Gravitational redshift (weak field): } \mathbf{M} \approx \mathbf{z}_{\text{grav}} \mathbf{c}^2 \mathbf{r} / \mathbf{G}$$

$$\text{Lensing deflection (thin-lens scaling): } \mathbf{M} \approx \mathbf{a} \mathbf{c}^2 \mathbf{b} / (4\mathbf{G})$$

On this view, these relations are not introduced as independent laws; rather, they are treated as distinct observational routes to inferring the same source strength that determines $\mathbf{A}$ in the relevant regime.

Principle (STAM). Spacetime admits a dimensionless accumulation field $\mathbf{A}$ sourced by matter–energy. Spatial gradients of $\mathbf{A}$ define the local gravitational acceleration field, while line integrals of $\mathbf{A}$ along propagation paths define traversal delay and redshift–distance

excess. The threshold $\mathbf{A}=1$ marks a critical propagation boundary associated with horizon behavior.

Spacetime possesses a dimensionless accumulation field, $\mathbf{A}$. Matter-energy sources this accumulation. Gradients of $\mathbf{A}$ produce gravitational acceleration. Path integrals of $\mathbf{A}$ produce traversal delay and redshift-distance excess. The threshold $\mathbf{A} = 1$ marks a critical propagation boundary.

STAM therefore proposes a compact unifying interpretation:

Unifying correspondences (summary).

Gravitation: local gradient of $\mathbf{A}$

Propagation delay: path integral through $\mathbf{A}$

Redshift–distance excess: accumulated traversal contribution

Inferred distance: geometric proxy + traversal excess

Horizon: critical accumulation threshold ($\mathbf{A}=1$)

Mass: source strength inferred from $\mathbf{A}$ via multiple observables

The model does not require these effects to be understood first as separate phenomena. Instead, it treats them as related consequences of spacetime accumulation.

6. Falsifiability, Scope Conditions, and Open Problems

- **Consistency with classical tests.** In regimes where $\mathbf{A}(\mathbf{r})=2GM/(\mathbf{c}^2\mathbf{r})$ is assumed, the model should reproduce (within stated approximations) Newtonian acceleration and the functional form of weak-field Shapiro delay; any departures should be parameterized and confronted with Solar System bounds.
- **Gravitational redshift mapping.** The model should specify an explicit relation between $\mathbf{A}$ and gravitational redshift (e.g., via clock-rate or frequency-shift prescriptions) and show agreement with weak-field measurements.
- **Lens and dynamics closure.** If $\mathbf{A}$ is fundamental, the same field that predicts acceleration should also predict light deflection and time delay without introducing inconsistent normalization factors across observables.
- **Cosmological prediction with locked parameters.** The phenomenological form of $\mathbf{D}_{\text{geo}}$, $\mathbf{D}_{\text{excess}}$, and $\mathbf{A}_{\text{path}}(z)$ should be derived from independent assumptions (or fixed by one dataset) and then tested on independent distance indicators (e.g., SNe Ia subsets, BAO-inferred scales, strong-lens time delays) without retuning.
- **Horizon regime dynamics.** The model requires a concrete dynamical prescription for the $\mathbf{A}\geq 1$ regime (coordinate-invariant statement, causal structure, and relation—if any—to Schwarzschild/Kruskal descriptions).

STAM's primary motivation is theoretical economy: a single scalar variable $\mathbf{A}$ is used to organize a set of gravitational and cosmological observables that are typically introduced through distinct formalisms. The principal unresolved task is to replace phenomenological cosmological prescriptions (e.g., the proposed $\mathbf{A}_{\text{path}}(z)$ form) with an independent derivation and to evaluate locked-parameter predictions against multiple, independent datasets.

In short:

Summary statement. STAM proposes that a dimensionless spacetime accumulation field can serve as a common explanatory and inferential substrate for gravitation, propagation delay, redshift–distance phenomenology, and horizon criteria.